

4. Business Requirements Document

❖ Section 1: Executive Summary

Pharma-Pulse is a procurement intelligence system designed to provide district- and state-level public health procurement officers in India with 8–12 weeks' warning of generic drug shortage risk.

Current procurement practice is entirely reactive. Shortage signals available in public data (API price movements, regulatory backlogs, disease outbreak patterns, supplier concentration) go unread until a hospital shelf is empty. By that point, the standard procurement lead time of 5–6 months makes timely recovery operationally impossible.

This document defines the business requirements for Pharma-Pulse Version 1.0, a working proof of concept targeting India's public health procurement segment, specifically, district and state procurement agencies that manage essential generic medicines for government health facilities.

The system is not intended to replace existing hospital inventory management software. It operates one layer upstream, at the procurement decision level, providing forward-looking intelligence that no existing system currently provides.

❖ Section 2: Business Objectives

ID	Business Objective	Success Measure
BO-01	Provide procurement officers with actionable shortage risk intelligence at least 8 weeks before projected onset	Average lead time of model alerts vs actual shortage onset (target 8+ weeks)
BO-02	Reduce reactive procurement decisions by enabling pre-emptive tender initiation	Reduction in emergency procurement events in pilot facilities
BO-03	Aggregate fragmented public data signals into a single accessible intelligence source	All six data sources integrated and refreshed on a weekly cycle
BO-04	Deliver insights in a format usable by non-technical procurement staff	Procurement officer can interpret dashboard without training in under 5 minutes
BO-05	Build a system that is transparent and auditable; every risk score must be explainable.	Any risk score can be decomposed to its contributing features within 2 clicks
BO-06	Demonstrate that this problem can be solved with publicly available data at no cost	All data sources used are free and publicly accessible

❖ Section 3: Stakeholder Requirements

SR: District Procurement Officer (Primary User) Source: Secondary research IJMR paper, CMHO Phalodi tender, WHO 2016 report

ID	Requirement	Priority
SR-01	I need to know which drugs are at risk of running out before they actually run out	Critical
SR-02	I need to know why a drug is at risk, not just a number, but the actual reason	Critical
SR-03	I need to know what to do about it, which supplier to contact, how much stock to build	High
SR-04	I need the information in a format I can show my senior officer to justify early procurement	High
SR-05	I need to be able to check the status quickly; I do not have time for a complex system	High
SR-06	I need to trust the system- if it raises too many false alarms, I will stop using it.	Medium

SR: State Health Agency Officer (Primary User) Source: Secondary research, TNMSC procurement documentation, WHO 2016

ID	Requirement	Priority
SR-07	I need a state-level view of which districts and drug categories are most at risk	Critical
SR-08	I need to be able to compare risk across states to prioritise where to act first	High
SR-09	I need to understand seasonal patterns so I can plan buffer stock before peak demand	High
SR-10	I need the system to flag when a supplier concentration risk is building before it becomes a crisis	High

SR: Generic Drug Manufacturer (Stakeholder) Source: WHO 2016, IJMR paper 2025

ID	Requirement	Priority
SR-11	Visibility into demand signals from public procurement would help us plan production cycles more accurately	Low

❖ Section 4: Functional Requirements

1. Data Ingestion

Requirement	The system shall ingest data from six defined sources on a weekly refresh cycle
Acceptance Criteria	All six data sources are loaded, cleaned, and processed without manual intervention within the weekly refresh window

2. Risk Score Calculation

Requirement	The system shall calculate a shortage risk score between 0 and 100 for each drug-region combination in the defined drug list
Acceptance Criteria	A risk score is produced for every drug in the defined list for every Indian state in the model, updated with each weekly refresh

3. Risk Score Explainability

Requirement	The system shall display the top contributing feature for each risk score alongside the score itself
Acceptance Criteria	Every drug risk score on the dashboard shows the primary driver — for example "API price up 31%" or "Supplier HHI 0.74" — without requiring the user to drill down

4. Regional Heatmap

Requirement	The system shall display a state-level shortage risk heatmap covering all major Indian states
Acceptance Criteria	Heatmap renders correctly showing colour-coded risk levels for all states. User can identify highest-risk states at a glance without reading any numbers

5. Drug-Level Drilldown

Requirement	The system shall allow the user to select any drug and view its full risk profile — score, trend, contributing features, and supplier reliability
Acceptance Criteria	Clicking any drug on the dashboard produces a detailed view showing current score, 12-week trend, top three risk drivers, and at least one alternative supplier with reliability score

6. Procurement Recommendation Engine

Requirement	The system shall generate a plain-language procurement recommendation for every drug flagged at high or critical risk
Acceptance Criteria	Each high or critical risk drug shows a recommendation stating: urgency level, recommended buffer stock increase, and at least one alternate supplier ranked by reliability score

7. Scenario Simulator

Requirement	The system shall provide an interactive scenario simulator allowing the user to adjust key upstream variables and see the projected impact on shortage risk
Acceptance Criteria	Simulator includes at minimum three adjustable variables — API export restriction percentage, demand surge percentage, and buffer stock level. Risk score updates in response to slider movement.

8. Alert Threshold Configuration

Requirement	The system shall categorise risk scores into four alert levels with defined thresholds
Acceptance Criteria	Risk scores are consistently categorised as Critical (80–100), High (60–79), Medium (40–59), and Low (0–39). Colour coding matches across all dashboard views.

❖ Section 5: Non-Functional Requirements

	Requirement	Acceptance Criteria	Priority
Usability	A procurement officer with no data science background must be able to interpret the dashboard without training	Comprehension test passed in under 5 minutes by a non-technical user	Must have
Refresh cadence	System must produce updated outputs on a weekly cycle aligned to IDSP bulletin publication	New risk scores available within 48 hours of weekly data refresh	Must have
Explainability	No risk score may be presented without an accompanying plain-language explanation of its primary driver	100% of scores on dashboard show primary driver label	Must have
Transparency	All data sources, feature weights, and model assumptions must be documented and accessible	Full Data Dictionary published alongside system	Must have
Accuracy	Model backtest accuracy against known shortage events must exceed 75%	Precision score above 0.75 on historical validation set	Should have

Portability	Dashboard must be openable by a standard government office computer without specialist software	Runs in Power BI Desktop or Tableau Public (both free tools)	Must have
Scalability	Architecture must be extensible to additional drug categories and states without fundamental redesign	New drug or state can be added by updating data dictionary and rerunning Python pipeline	Should have
Data cost	All data sources used must be free and publicly accessible	Zero cost data architecture confirmed: all six sources are public	Must have

❖ Section 6: MoSCoW Prioritisation

Must Have

- Risk score calculation for defined drug list across all states
- Risk score explainability
- Regional heatmap
- Drug-level drilldown with trend and supplier information
- Procurement recommendation for high and critical drugs
- Four-level alert threshold system
- Model back-test validation report
- Full Data Dictionary documentation

Should Have

- Scenario simulator with three adjustable variables
- 12-week rolling trend sparklines
- Seasonal pattern layer activating pre-monsoon and monsoon alerts
- Supplier reliability scoring with alternate supplier ranking

Could Have (Future versions)

- Automated PDF extraction from IDSP bulletins replacing manual download
- Direct CDSCO API integration if government data infrastructure improves
- State procurement officer login and personalised state-level view
- Mobile-responsive dashboard version
- Email alert system for threshold breaches

Won't Have

- Real-time data streaming: weekly refresh is the correct cadence for this problem
- Private sector hospital data integration: scope is public health procurement only
- Manufacturer-facing portal: out of scope for procurement intelligence system
- Prescription or patient-level data: privacy and access constraints make this infeasible

❖ Section 7: Assumptions and Constraints

ID	Assumption	Risk if Wrong
A-01	All six data sources remain publicly accessible throughout the project	Medium- CDSCO portal accessibility has been inconsistent historically
A-02	Weekly data refresh cadence is sufficient for the procurement decision horizon	Low- shortage development timelines of 8–12 weeks confirm this
A-03	World Bank commodity indices are a valid proxy for upstream API cost pressure	Low- validated by FOPE warning correlation
A-04	IDSP outbreak reporting is sufficiently complete to generate meaningful signals	Medium- under-reporting in peripheral states documented in DQA
A-05	The defined drug list of six core drugs is representative of highest-risk essential medicines	Low- independently validated by FDA, WHO, and TMH primary research

ID	Constraint	Impact
C-01	All data sources must be free and publicly available	Limits direct API price data access
C-02	No Cloud data ingestion	Limits real-time streaming capability: weekly refresh is maximum cadence
C-03	System scope is limited to public health procurement	Limits validation dataset size (private hospital data unavailable)
C-04	Project timeline is one academic year	Limits model training data to available historical records (3–4 years for GeM data)